ULTRAVIOLET RADIATION

UVR represents approximately 5 percent of terrestrial solar radiation and is officially divided into UVA (315 to 400 nm), UVB (280 to 315 nm), and UVC (100 to 280 nm), though other definitions are widely used (e.g., a UVB/UVA division at 320 nm). UVA is sometimes subdivided into UVAI (340 to 400 nm) and UVAII (315/320 to 340 nm). Sealevel solar UVR is approximately 95 percent to 98 percent UVA and 2 percent to 5 percent UVB. UVC is completely absorbed by stratospheric ozone (O_3) that also attenuates UVB. A given solar UVR spectrum varies with the solar zenith angle (the angle between an imaginary perpendicular line and a line from its base to the sun), which is dependent on time of day, season, and latitude. The highest UVB content is found when the sun is directly overhead with the shortest path (e.g., noon at the equator at high altitude).

Measurement of Human Exposure to Ultraviolet Radiation

The measurement of UVR exposure is called dosimetry and can be done physically or biologically. Accurate physical dosimetry, in which dose rate (irradiance) is measured as $\mathrm{W/cm}^2$ and exposure dose as $\mathrm{J/cm}^2$ [$\mathrm{W/cm}^2$ × time (seconds)], is technically demanding, and a biologic approach is more commonly used. This is based on the individual minimal erythemal dose (MED), which is the lowest dose required to produce a just perceptible erythema (sunburn) 24 hours later. In general, individuals with higher skin types have higher MEDs. However, there is considerable MED overlap between skin phototypes.

Most human UVR exposure is from sunlight, but the use of tanning devices is increasingly popular. Solar UVR exposure may be intentional or inadvertent and depends on specific behaviors and time spent outdoors, as well as photoprotection strategies.

ACUTE AND CHRONIC EFFECTS OF ULTRAVIOLET RADIATION ON THE SKIN

AT A GLANCE

- Acute and chronic responses to ultraviolet radiation (UVR) depend on skin phototype.
- The major acute clinical effects are inflammation, tanning, and immunomodulation.
- Basal cell carcinoma, squamous cell carcinoma, malignant melanoma, and photoaging are the chronic effects of UVR.
- The majority of effects of solar UVR on the skin are the result of its small UVB content.

Action Spectroscopy

In any skin response to UVR, it is important to know which wavelengths are primarily responsible. This relationship is known as the action spectrum and is fundamentally based on the UVR-absorbing properties of the molecule (chromophore) that initiates the effect (see Chap. 88).